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# Sensitivity of Prescribed-Potential Corrosion

Analytical steady states, resistance shares, and parameter trends

Corrosion Modeling Project

## Principal result

At the thermal-vacancy reference, steady removal is almost equally sensitive to fractional changes in Cr diffusivity, vacancy reference and inverse transport length; voltage has negligible influence. This ranking reverses when bulk transport becomes fast enough for the interface to limit the rate.

## Scope and the quantity being ranked

The response is the **steady Cr-equivalent removal rate**, expressed in  $\mu\text{m}/\text{yr}$ . It is the dissolved Cr molar flux divided by molar site density, not wall recession or depletion depth. The study uses the existing *linear-surface, constant-coefficient analytical benchmark*: full thermodynamic Hessian and full kinetic mobility (PTK), with a secondary thermodynamic-only (PT) comparison. No new MOOSE simulations are used. The nonlinear activity-prefactor alternative in the separate Butler–Volmer report is not substituted into this benchmark.

The interval runs from the corroding surface  $x = 0$  to a fixed alloy reservoir at  $x = L$ . Initial/far occupied-site Fe:Ni:Cr ratios are preserved when the vacancy reference changes. In this geometry  $L$  is a transport length; interpreting it as a grain size assumes the specified far-boundary reservoir represents that grain.

$$\mathbf{D} = \mathbf{B}\mathbf{H}, \quad \mathbf{a} = \mathbf{D}^{-1}\mathbf{s}, \quad \mathbf{q}(x) = \mathbf{q}_b - (L - x)\mathbf{a}r, \quad \mathbf{s} = (0, 1, -1)^T. \quad (1)$$

Here  $\mathbf{q} = (x_{\text{Ni}}, x_{\text{Cr}}, x_v)^T$ ;  $\mathbf{H}$  and  $\mathbf{B}$  are rebuilt at each trial reference, then held spatially constant. Cr dissolves, Ni has zero interfacial flux, and vacancies enter the alloy. With  $c_b = x_{\text{Cr},b}$  and  $c_s = x_{\text{Cr},s}$ ,

$$r = hc_s, \quad h = \frac{j_* \sinh u}{FC_{\text{site}}c_b}, \quad u = \frac{\eta}{V_T}, \quad V_T = \frac{RT}{F}. \quad (2)$$

**Equation meaning.** Actual  $\eta$  is fixed, so a changing equilibrium potential does not reduce the imposed voltage factor. This is the retained symmetric two-electron interface normalization, with  $j_* = 5 \text{ A}/\text{m}^2$  as a diagnostic input, not an experimental calibration.

$$r = \frac{c_b}{La_{\text{Cr}} + 1/h} = \frac{r_D r_I}{r_D + r_I}, \quad r_D = \frac{c_b}{La_{\text{Cr}}}, \quad r_I = hc_b. \quad (3)$$

$r_D$  and  $r_I$  are hypothetical limiting *capacities*, not two different realized fluxes: the actual bulk and interface fluxes are equal at steady state. A speed of 1 m/s corresponds to  $3.1536 \times 10^{13} \mu\text{m}/\text{yr}$  (365-day year).

## A fair sensitivity ranking

### Compare equal fractional changes

$$f_D = \frac{La_{\text{Cr}}}{La_{\text{Cr}} + 1/h} = \frac{r_I}{r_D + r_I}, \quad f_I = 1 - f_D = \frac{r_D}{r_D + r_I}. \quad (4)$$

$$\epsilon_p = \frac{\partial \ln r}{\partial \ln p}; \quad \frac{\Delta r}{r} \simeq \epsilon_p \frac{\Delta p}{p}. \quad (5)$$

A magnitude of one means a 1% input change produces approximately a 1% rate change locally. A negative sign indicates a decrease. This is not a probability-based global sensitivity analysis; no distributions or parameter uncertainties have been assumed.

$$\boxed{\epsilon_L = -f_D, \quad \epsilon_{D_{\text{Cr}}^*} = f_D, \quad \epsilon_{S_v} = f_D g_S, \quad \epsilon_\eta = f_I u \coth u,} \quad g_S = \frac{\partial \ln r_D}{\partial \ln S_v}. \quad (6)$$

These formulas are exact for this PTK benchmark. Changing  $D_{\text{Cr}}^*$  changes the Cr tracer input and rebuilds its linked mobility entries, while Fe/Ni tracers remain fixed. Changing  $S_v = x_{v,b}/x_{v,\text{th}}$  rebuilds both the Hessian and mobility. The thermal normalization in the mobility stays fixed.

$$\mathbf{B}^{-1}\mathbf{s} = (0, 1/w_{\text{Cr}}, 0)^T, \quad w_{\text{Cr}} = \frac{D_{\text{Cr}}^* c_b S_v}{RT}, \quad a_{\text{Cr}} = \frac{(\mathbf{H}^{-1})_{\text{CrCr}}}{w_{\text{Cr}}}. \quad (7)$$

This explains the exact Cr-tracer elasticity in Eq. (6); it does not remove thermodynamic coupling or describe transient cancellation. Numerically  $g_S$  falls only from 1.000 to 0.962 across  $S_v = 1\text{--}10^6$ .

| Operating state          | $f_D$  | $\epsilon_L$ | $\epsilon_D$ | $\epsilon_{S_v}$ | $\epsilon_\eta$       |
|--------------------------|--------|--------------|--------------|------------------|-----------------------|
| Thermal baseline         | 1.000  | −1.000       | 1.000        | 1.000            | $4.58 \times 10^{-7}$ |
| Equal resistance         | 0.500  | −0.500       | 0.500        | 0.500            | 0.657                 |
| Interface dominated      | 0.0279 | −0.0279      | 0.0279       | 0.0279           | 1.276                 |
| 83% diffusion resistance | 0.831  | −0.831       | 0.831        | 0.831            | 1.013                 |

Table 1: All states use  $S_v = 1$ . Baseline:  $L = 100 \mu\text{m}$ ,  $M_{\text{Cr}} = 1$ ,  $\eta = V_T$ . Other rows use  $L = 1 \mu\text{m}$ :  $M_{\text{Cr}} = 28691.47$ ,  $\eta = V_T$ ;  $M_{\text{Cr}} = 10^6$ ,  $\eta = V_T$ ; and  $M_{\text{Cr}} = 10^6$ ,  $\eta = 6V_T$ , respectively.  $M_{\text{Cr}} = D_{\text{Cr}}^*/D_{\text{Cr},0}^*$  is the tracer multiplier.

Here  $w_{\text{Cr}}$  is the dimensional Cr exchange mobility;  $M_{\text{Cr}}$  is its tracer-input multiplier. **Sensitivity is not rate control.** The last row is diffusion dominated, yet equal-percent voltage changes have the largest local effect because  $u \coth u \simeq 6$ . Resistance share alone does not rank arbitrary input sensitivities.

## Length and Cr diffusivity

Unless specified otherwise, calculations use  $T = 973.15$  K,  $V_T = 83.8596$  mV,  $D_{Cr,0}^* = 8.77488 \times 10^{-20}$  m<sup>2</sup>/s,  $x_{v,th} = 2.91878 \times 10^{-8}$  and  $C_{site} = 10^5$  mol/m<sup>3</sup>. The defaults are  $L = 100$   $\mu$ m,  $S_v = M_{Cr} = 1$  and  $\eta = V_T$ .

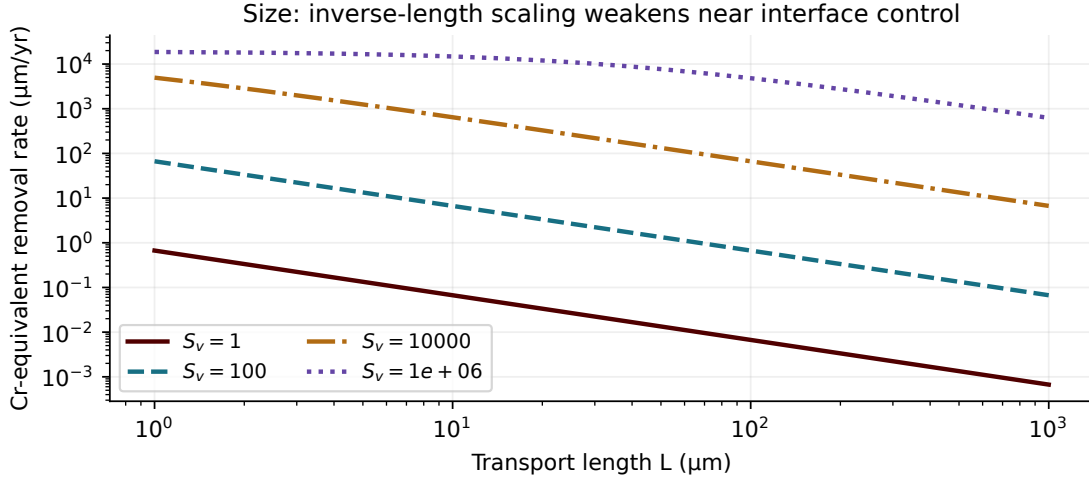


Figure 1: Length sweep at  $\eta = V_T$ ,  $M_{Cr} = 1$ . At the thermal reference, increasing  $L$  from 1 to 1000  $\mu$ m lowers the rate from 0.66936 to  $6.69383 \times 10^{-4}$   $\mu$ m/yr. High vacancy references increase supply and weaken the inverse-length trend at small  $L$ .

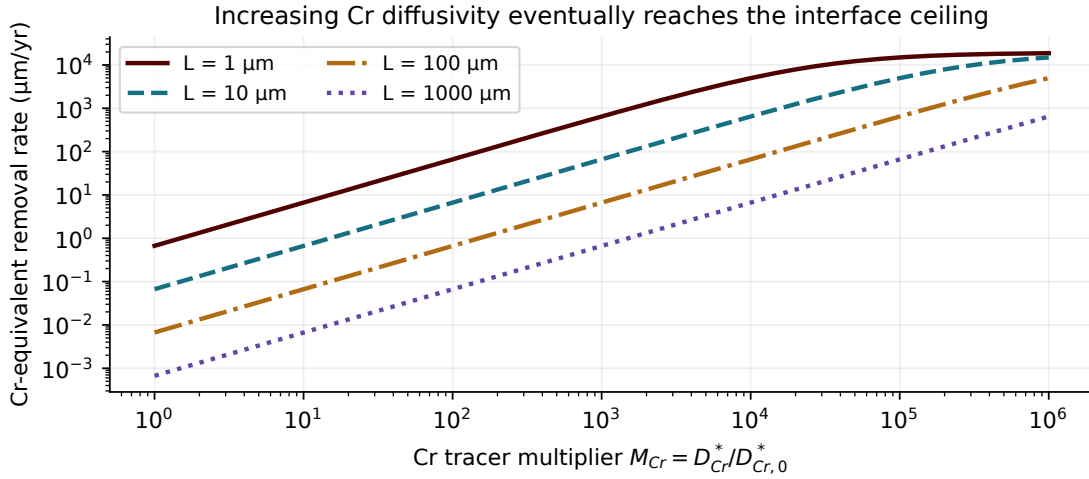


Figure 2: Cr tracer sweep at  $\eta = V_T$ ,  $S_v = 1$ . Short paths reach the interface capacity  $r_I = 19205.6$   $\mu$ m/yr first. This large capacity comes from the diagnostic reaction scale; the much smaller actual rate is the harmonic combination in Eq. (3).

**Interpretation.** Reducing  $L$  and raising  $D_{Cr}^*$  are exactly opposite logarithmic changes to the same diffusion resistance in PTK. Consequently  $\epsilon_D = -\epsilon_L$  at every point, not just at the baseline. This is a statement about one-dimensional transport length, not an independently modeled grain-boundary network.

## Vacancy reference and prescribed overpotential

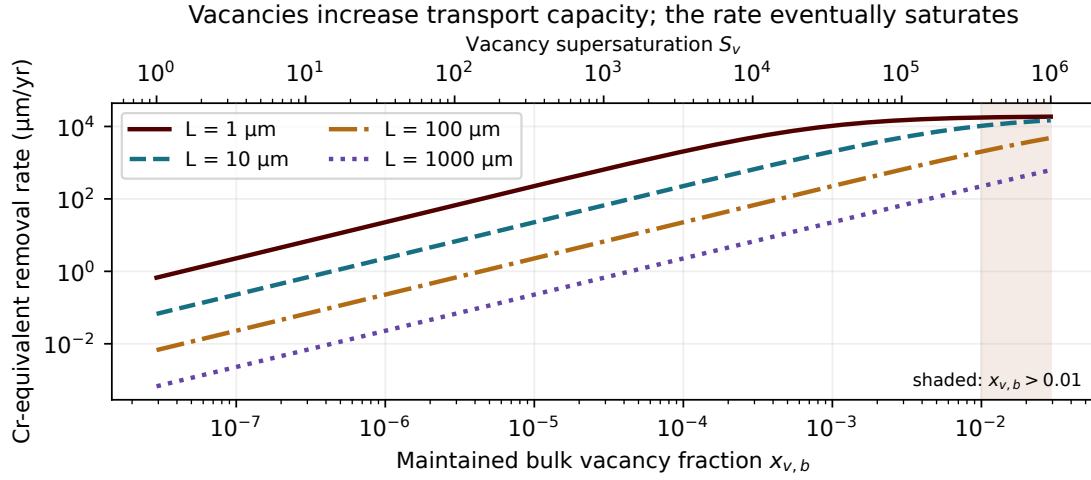


Figure 3: Vacancy sweep at  $\eta = V_T$ ,  $M_{Cr} = 1$ . The upper axis gives  $S_v$ ; the lower axis gives the actual maintained vacancy fraction.  $S_v = 10^6$  means  $x_{v,b} = 0.02919$ . Shading marks  $x_{v,b} > 0.01$ , where a void-free alloy interpretation is particularly inappropriate; it is not a validated onset threshold.

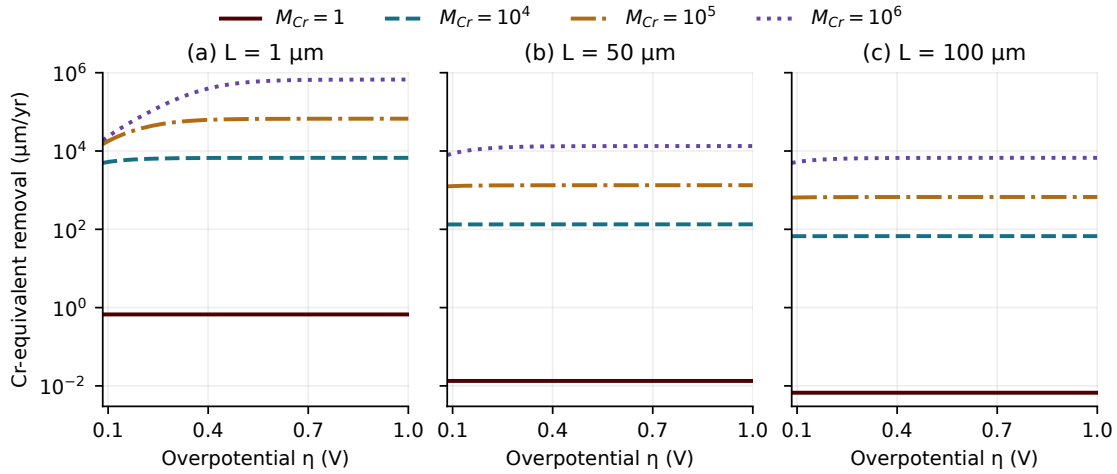


Figure 4: Voltage sweeps at  $L = 1, 50, 100 \mu\text{m}$  with  $S_v = 1$ , identical axes and the same tracer multipliers in every panel. Larger lengths lower the supply ceiling and weaken the voltage response. At unmodified Cr diffusivity all three curves are nearly flat.

**Different physical actions.** Increasing the maintained vacancy reference enhances vacancy-mediated transport in this benchmark. Increasing *actual prescribed*  $\eta$  accelerates the interface. Neither sweep represents the vacancy-induced loss of overpotential in a mixed-potential model. Here the reference vacancy fraction is an input; the surface vacancy fraction follows from Eq. (1).

## When the ranking changes

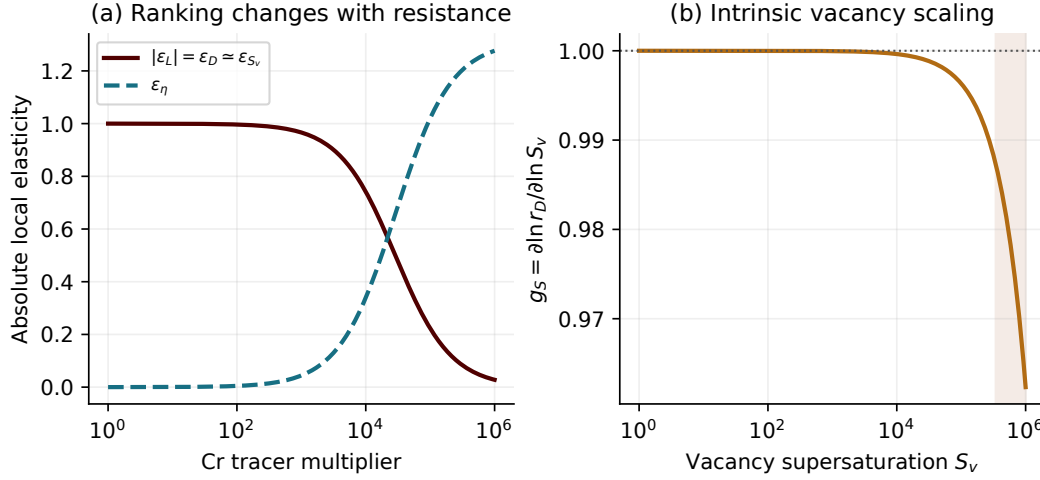


Figure 5: (a) At  $L = 1 \mu\text{m}$ ,  $S_v = 1$ ,  $\eta = V_T$ , increasing Cr mobility reduces the transport elasticities and increases voltage elasticity. The three transport curves coincide to plotting precision. (b)  $g_S$  measures the response of the diffusion capacity before its resistance weighting; composition and Hessian changes make it slightly smaller than one at large  $S_v$ .

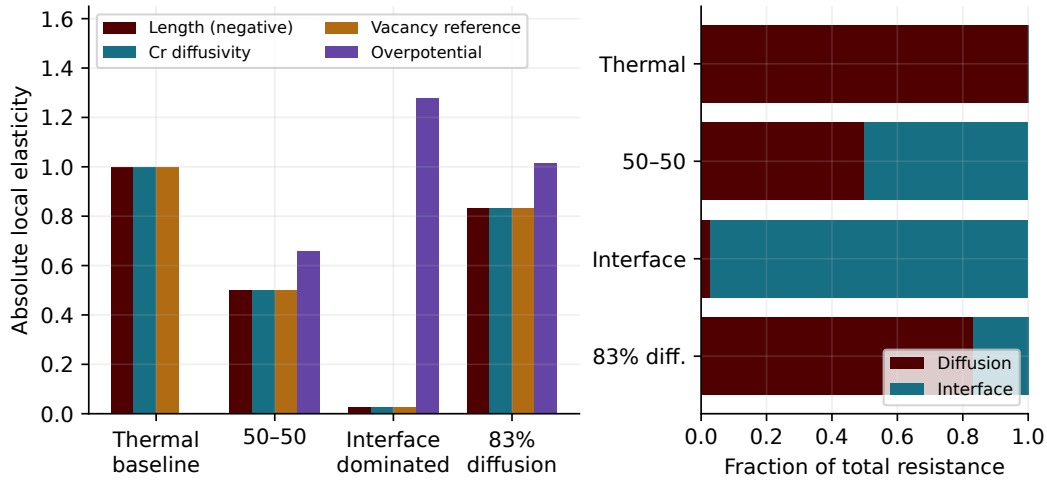


Figure 6: The four states in Table 1. Left: absolute elasticities, with length always negative. Right: resistance shares. A high voltage elasticity in the final case does not imply that the interface supplies most of the resistance.

$$\epsilon_\eta > \epsilon_D \iff f_D < \frac{u \coth u}{1 + u \coth u}. \quad (8)$$

At  $\eta = V_T$  the crossover is  $f_D = 0.568$ ; at  $6V_T$  it is 0.857. A sensitivity ranking and a statement about the rate-limiting resistance therefore answer different questions.

## Joint effect of length and Cr diffusivity

Shorter paths and faster Cr transport raise the rate

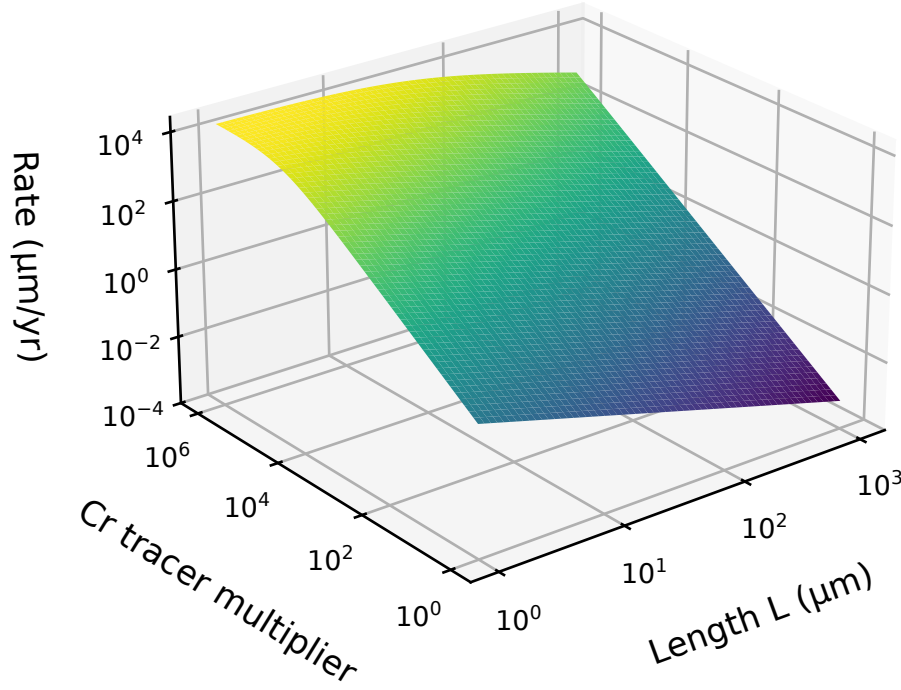


Figure 7: PTK steady rate at  $S_v = 1$ ,  $\eta = V_T$ . Both horizontal axes and the vertical axis are logarithmic (ticks show untransformed values). Color reinforces the vertical rate. Every surface contains  $61 \times 61$  directly evaluated states.

$$r \simeq r_D = \frac{D_{\text{Cr}}^* S_v c_b^2}{RT L (\mathbf{H}^{-1})_{\text{CrCr}}} \propto \frac{D_{\text{Cr}}^* S_v}{L} \quad \text{when the reference thermodynamic factor changes little.} \quad (9)$$

The sloping region follows this supply law. The flattened corner appears because the interface cannot consume the supplied Cr any faster at the chosen voltage. Raising  $D_{\text{Cr}}^*$  beyond that point mainly raises surface Cr toward the reservoir value, while the rate approaches  $r_I$ .

At  $L = 1 \mu\text{m}$  and the thermal vacancy reference, the Cr multipliers giving 90% diffusion resistance, equal resistance and 90% interface resistance are approximately **3188**, **28691** and **258223**, respectively. At  $100 \mu\text{m}$  each threshold multiplier is 100 times larger. Thus the longer specimen remains more transport limited over the same diffusivity range.

## Joint effect of length and vacancy reference

Vacancy enhancement is strongest before the interface plateau

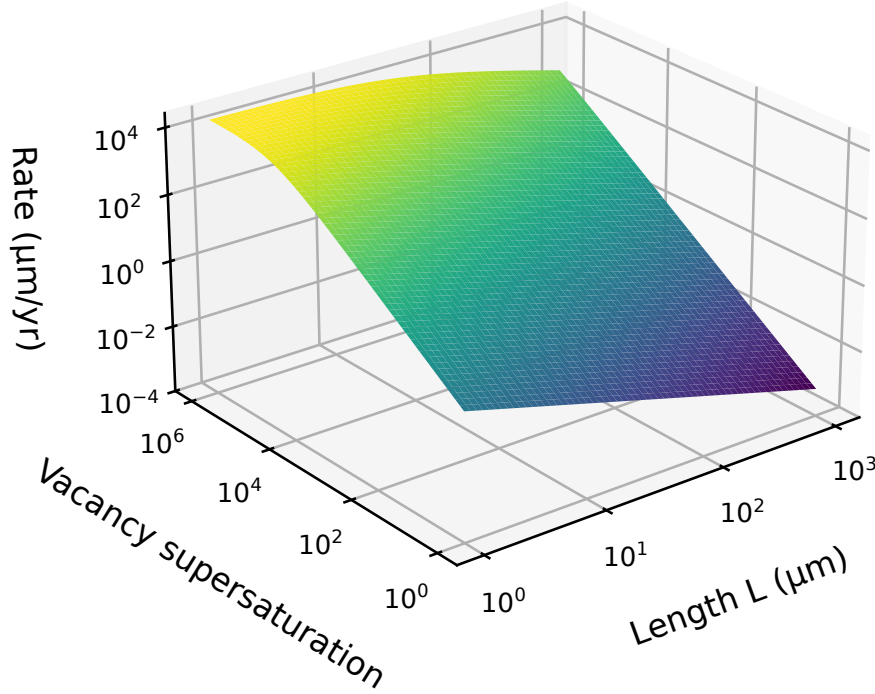


Figure 8: PTK steady rate at  $M_{Cr} = 1$ ,  $\eta = V_T$ . Axes are logarithmic. The segment  $S_v > 3.4261 \times 10^5$  has a maintained vacancy fraction above 1%; it is retained to show mathematical trends, not to predict a stable pore-free alloy.

The similarity to the preceding surface is a consequence of the retained mobility law, not a new fit. For dilute vacancies, raising  $S_v$  multiplies the exchange mobilities almost uniformly while changing the relevant thermodynamic compliance only slightly. Across the full range, however, metal site fractions decrease and the Hessian changes, so vacancy and Cr-tracer enhancement are not exactly interchangeable.

$$L_{50} = \frac{1}{ha_{Cr}}, \quad L_{90D} = 9L_{50}, \quad L_{90I} = L_{50}/9. \quad (10)$$

Increasing  $S_v$  or Cr diffusivity increases  $L_{50}$ : a longer transport path is required before diffusion supplies most resistance. Increasing voltage has the opposite effect because it reduces interface resistance. At the baseline  $L_{50} = 3.4854 \times 10^{-5} \mu\text{m}$ , far below the studied lengths. At  $S_v = 10^6$  it is about  $33.58 \mu\text{m}$ .

## Joint effect of voltage and Cr diffusivity

Voltage matters when bulk supply becomes sufficiently fast

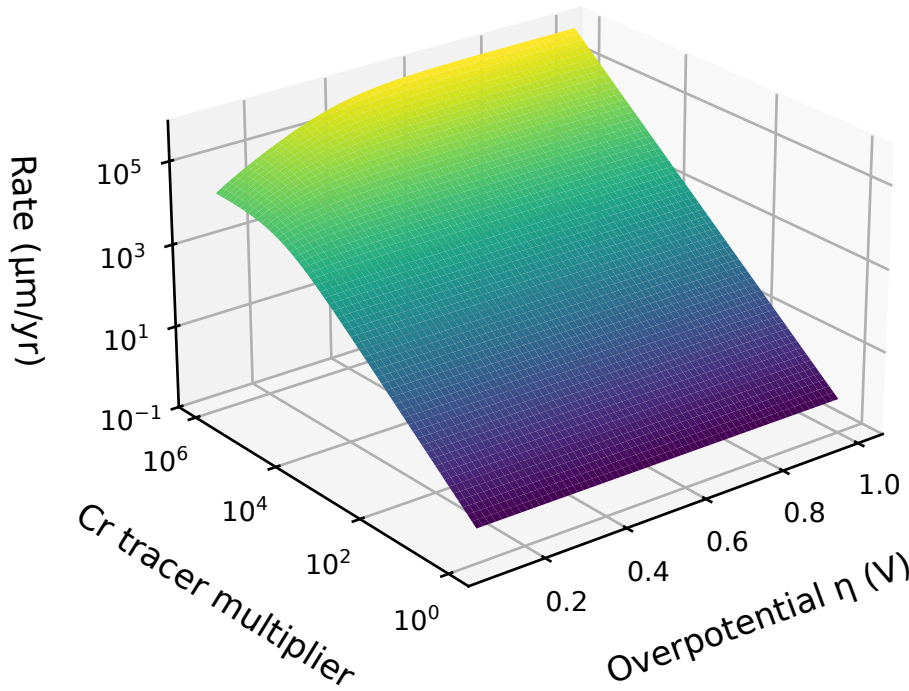


Figure 9: PTK steady rate at  $L = 1 \mu\text{m}$ ,  $S_v = 1$ . Voltage is linear in volts; diffusivity multiplier and rate axes are logarithmic. The surface rises with voltage only while appreciable interface resistance remains.

With  $M_{\text{Cr}} = 1$  and  $L = 100 \mu\text{m}$ , increasing voltage from  $V_T$  to 1 V increases rate by only **0.0000349%**. The diffusion capacity is already almost fully used. A very large change in the interface kinetic factor can therefore produce a negligible change in total steady removal.

At  $L = 1 \mu\text{m}$  and  $M_{\text{Cr}} = 10^6$ , by contrast, the state at  $V_T$  has 97.2% interface resistance. Its voltage elasticity is 1.276. At  $6V_T$ , diffusion supplies 83.1% of resistance, yet the voltage elasticity remains 1.013 versus 0.831 for Cr diffusivity. Increasing voltage further drives the rate toward its fixed supply ceiling, and ultimately makes voltage sensitivity small again.

**What the large rates mean.** These extreme tracer multipliers explore how the analytical model changes regime. They are not measured diffusivity ranges or credible wall-loss predictions. The multiplier  $10^6$  corresponds to  $D_{\text{Cr}}^* = 8.77 \times 10^{-14} \text{ m}^2/\text{s}$ ; its physical origin would require separate evidence.

## Joint effect of vacancies and Cr diffusivity

Vacancy and Cr mobility enhancement can become indistinguishable in rate

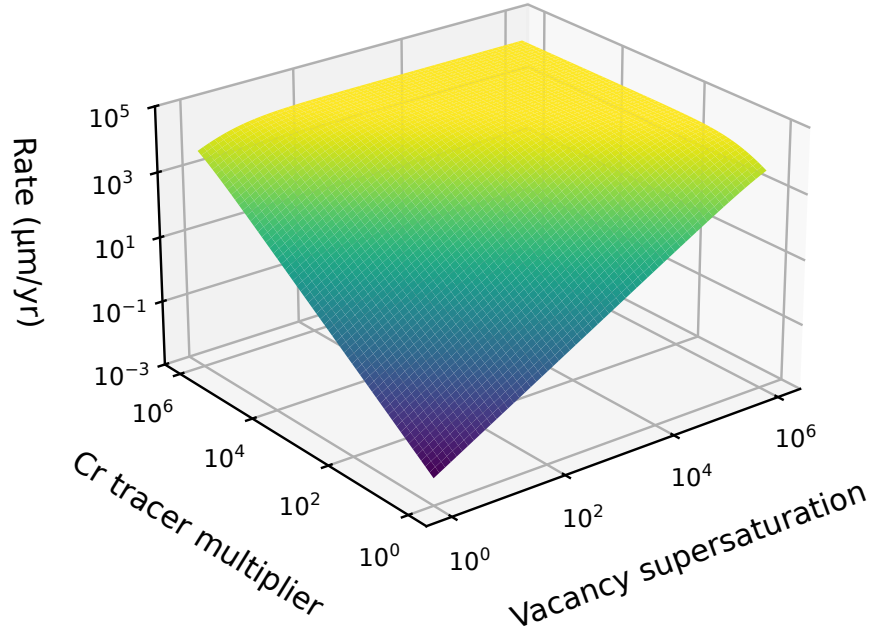


Figure 10: PTK steady rate at  $L = 100 \mu\text{m}$ ,  $\eta = V_T$ . All axes are logarithmic. In the rising region, either transport enhancement increases rate; at high combined enhancement the surface flattens toward the common interface capacity. The high-vacancy corner is extrapolative.

$$r \propto M_{\text{Cr}} S_v / L \implies M_{\text{Cr}} S_v \approx \text{constant along equal-rate contours at fixed } L. \quad (11)$$

This near-degeneracy has an important consequence: fitting only a stationary corrosion rate cannot reliably distinguish faster intrinsic Cr exchange from more available vacancies. Concentration profiles, time-dependent data or independent defect/mobility constraints would be needed to separate those explanations. Once interface controlled, even their product becomes weakly identifiable from the total rate.

The surface also shows why one-at-a-time rankings should not be treated as global. A vacancy increase can change the sensitivity to Cr diffusivity by moving the system into interface control; a diffusivity increase can similarly diminish vacancy sensitivity. Both inputs continue to enhance supply, but supply no longer determines most of the resistance.

## Steady polarization and Tafel representation

Express the same Cr-removal predictions as current for the retained two-electron reaction:

$$j = 2FC_{\text{site}}r = \frac{j_D j_I}{j_D + j_I}, \quad j_D = 2FC_{\text{site}}r_D, \quad j_I = 2j_* \sinh(\eta/V_T). \quad (12)$$

**Equation meaning.** Use  $r$  in m/s. With the retained site density,  $1 \mu\text{m}/\text{yr}$  corresponds to  $6.11906 \times 10^{-4} \text{ A}/\text{m}^2$ . Current and removal rate express the same response.

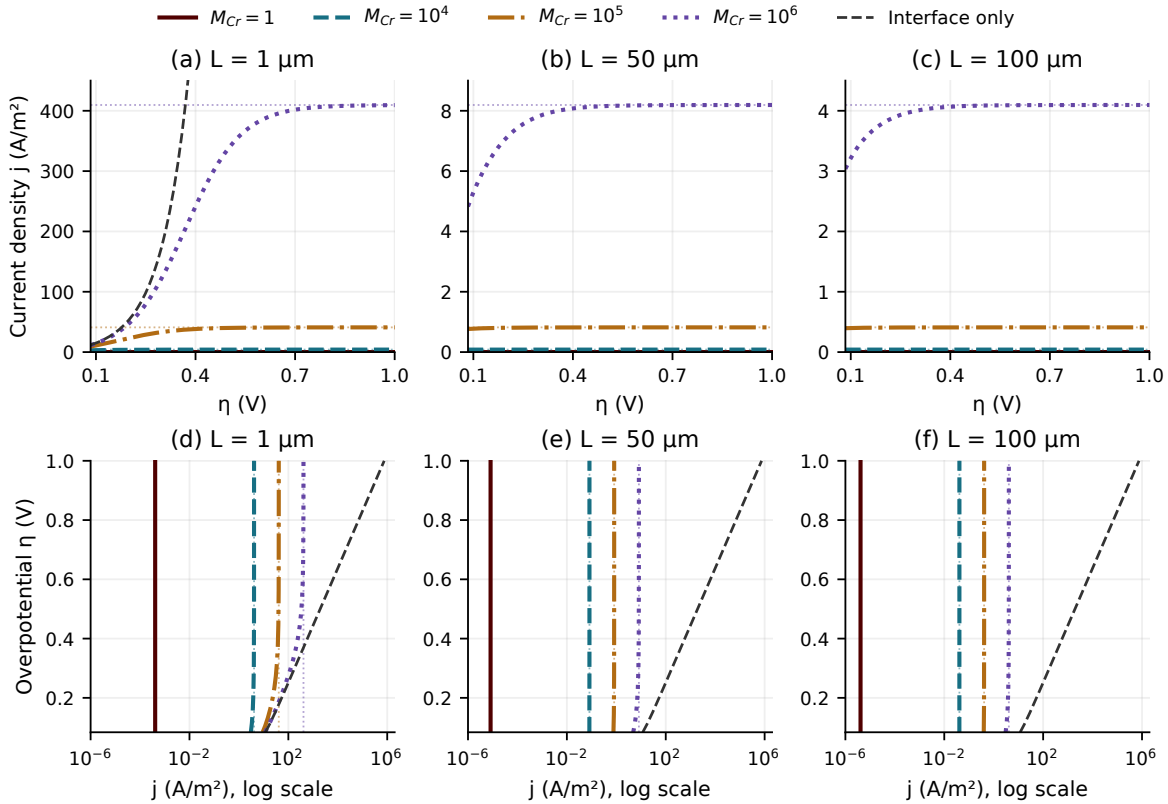


Figure 11: PTK at  $S_v = 1$  and  $700^\circ\text{C}$ . Columns:  $L = 1, 50, 100 \mu\text{m}$ ; top row: polarization; bottom row: Tafel representation. Colors retain  $M_{\text{Cr}} = 1, 10^4, 10^5, 10^6$ . **Top-row current scales differ**; bottom-row axes are identical. Black dashed: interface only at bulk composition (entirely above the current range in b–c). Faint colored dotted lines:  $j_D$ . Each curve has 301 stationary states from  $V_T$  to 1 V.

**Size comparison.** At  $M_{\text{Cr}} = 10^6$ , the diffusion limits are 409.60, 8.192 and 4.096  $\text{A}/\text{m}^2$  for 1, 50 and  $100 \mu\text{m}$ . Already at  $V_T$ , their diffusion-resistance shares are 2.79%, 58.93% and 74.15%. Larger lengths therefore saturate sooner, with less voltage response.

**Interpretation.** Steepening toward vertical in the Tafel panels reflects diffusion saturation, not a changed intrinsic charge-transfer slope. Each point is a separate steady state, not a finite-rate voltage sweep. The plotted variable is actual Cr overpotential  $\eta = E - E_{\text{eq}}$ ; changing surface composition also changes  $E_{\text{eq}}$ , so these are not directly experimental electrode-potential scans.

## Differential resistance and transport

Define the area-specific differential response to actual imposed overpotential,  $R_\eta = d\eta/dj$ . Hold  $L$ , the vacancy reference, tracer inputs and temperature fixed, and differentiate the *steady* current, allowing surface Cr to readjust. Equation (12) gives

$$R_\eta = \frac{V_T \tanh u}{j f_I} = \underbrace{\frac{V_T}{2j_* \cosh u}}_{R_{\eta,I}} \left(1 + \frac{j_I}{j_D}\right)^2, \quad u = \eta/V_T. \quad (13)$$

**Equation meaning.**  $R_\eta$  has units  $\Omega \text{ m}^2$ , unlike the transport resistances in Eq. (4).  $R_{\eta,I}$  is evaluated at bulk composition, not at the depleted surface.

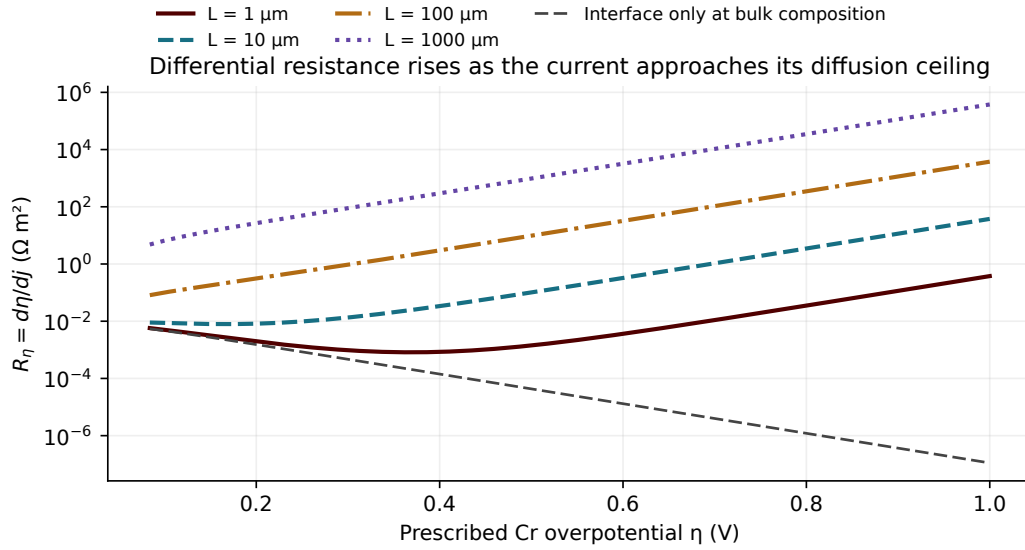


Figure 12: PTK at  $S_v = 1$ ,  $M_{Cr} = 10^6$  and  $700^\circ\text{C}$ , with length varied from 1 to  $1000 \mu\text{m}$ . The dashed reference omits depletion by evaluating the interface at bulk composition. For the shortest path, increasing voltage first lowers differential resistance, then increases it as the diffusion plateau is approached. Longer paths show a larger transport contribution.

The apparent Tafel slope follows directly:

$$b_\eta = \frac{d\eta}{d \log_{10} j} = (\ln 10) j R_\eta = \frac{(\ln 10) V_T \tanh u}{f_I}. \quad (14)$$

The interface-only anodic limit is  $(\ln 10) V_T = 0.1931 \text{ V/decade}$  for this symmetric two-electron law. It requires both  $u \gg 1$  and negligible depletion; near  $u = 1$ , the reverse branch still curves even the interface-only response. In the  $L = 1 \mu\text{m}$  case at  $6V_T$ , the coupled slope is  $1.144 \text{ V/decade}$  because diffusion supplies 83.1% of the resistance. At 1 V it is about  $356 \text{ V/decade}$ : current is almost saturated, not evidence of a changed transfer coefficient.

An experimental  $dE/dj$  also includes the changing equilibrium potential. Thus  $R_\eta$  is not directly an open-circuit polarization resistance or an impedance-derived charge-transfer resistance.

## Conclusions and reproducibility

### What can be concluded from these analytical solutions?

1. **Thermal-reference ranking:**  $|\epsilon_L| \simeq \epsilon_D \simeq \epsilon_{S_v} \simeq 1 \gg \epsilon_\eta$  for equal fractional input changes. Length has the opposite sign. The supply limit masks the intrinsic voltage dependence of Butler–Volmer kinetics.
2. **The ranking depends on the state.** Faster diffusion, elevated maintained vacancies or shorter paths reduce diffusion resistance. Voltage becomes more influential; transport and size elasticities decline.
3. **Resistance and sensitivity must be reported separately.** A large voltage elasticity does not establish interface control. Report resistance shares and elasticities together.
4. **Size and vacancy trends are conditional.** Fixed far-alloy reservoirs support nonzero steady supply. These trends do not transfer directly to closed grains or mixed potential.

| Sweep; other inputs at baseline | Start rate | End rate   | End/start             |
|---------------------------------|------------|------------|-----------------------|
| $L$ : 1 to 1000 $\mu\text{m}$   | 0.66936    | 0.00066938 | 0.00100003            |
| $M_{\text{Cr}}$ : 1 to $10^6$   | 0.00669383 | 4963.78    | $7.41545 \times 10^5$ |
| $S_v$ : 1 to $10^6$             | 0.00669383 | 4828.37    | $7.21316 \times 10^5$ |
| $\eta$ : $V_T$ to 1 V           | 0.00669383 | 0.00669383 | 1.000000349           |

Table 2: Rates in  $\mu\text{m}/\text{yr}$ . These finite-range changes are not a fair equal-uncertainty ranking because the input spans differ. Baseline:  $L = 100 \mu\text{m}$ ,  $S_v = M_{\text{Cr}} = 1$ ,  $\eta = V_T$ .

### Kinetic coupling and applicability

A 1488-state audit retained the Hessian and used diagonal mobility (PT). The largest sampled rate difference from PTK was 2.35%, at high vacancy reference; the thermal-baseline difference was  $8.72 \times 10^{-7}\%$ . Small stationary-rate differences do not establish negligible coupling effects on vacancy/Ni profiles or transients. The exact PTK identities are not imposed on PT.

The frozen coefficients approximate the nonlinear model, especially across strong depletion. High vacancy fractions require sink/void physics; tracer enhancement requires a physical mechanism. These steady-state rankings need not hold at early times. The figures are analytical parameter studies, not experimental uncertainty bands.

### Numerical checks and source trail

Verification covers the original 360-state engine comparison, 19594 sensitivity-figure states and 4816 electrochemical states. Added checks include 72 derivative identities and 39 resolved finite-difference comparisons, agreeing within  $1.3 \times 10^{-7}$  relative. Full records are in `QA/`.

`Source/` retains the original inputs, reference engine, Python calculations and editable report. `Data/` contains every plotted value. Original line studies use 181 points, electrochemical curves 301, and surfaces  $61 \times 61$  points. The complete launch also generates the new electrochemical figures and checks. See `README.md`, the package map and copy-safe `Launch_Commands.txt`.